# CM3121 – Advanced Organic Chemistry

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## Cycloadditions (part 1)

# Overview

- Introduction to pericyclic reactions, cycloaddition, and Diels–Alder reaction
- Mechanism of Diels–Alder reaction
- Kinetics of Diels–Alder reaction
- Stereochemistry of Diels–Alder reaction
- Frontier molecular orbitals and Woodward–Hoffmann rules

# Types of reactions based on mechanisms

**(1) Polar mechanism**

- Unsymmetrical heterolytic bond breaking or heterogenic bond making
- Electron rich sites in one molecule react with electron poor sites in another molecule involve the donation of an electron pair from a nucleophile to an electrophile (ionic intermediates)

**(2) Radical mechanism**

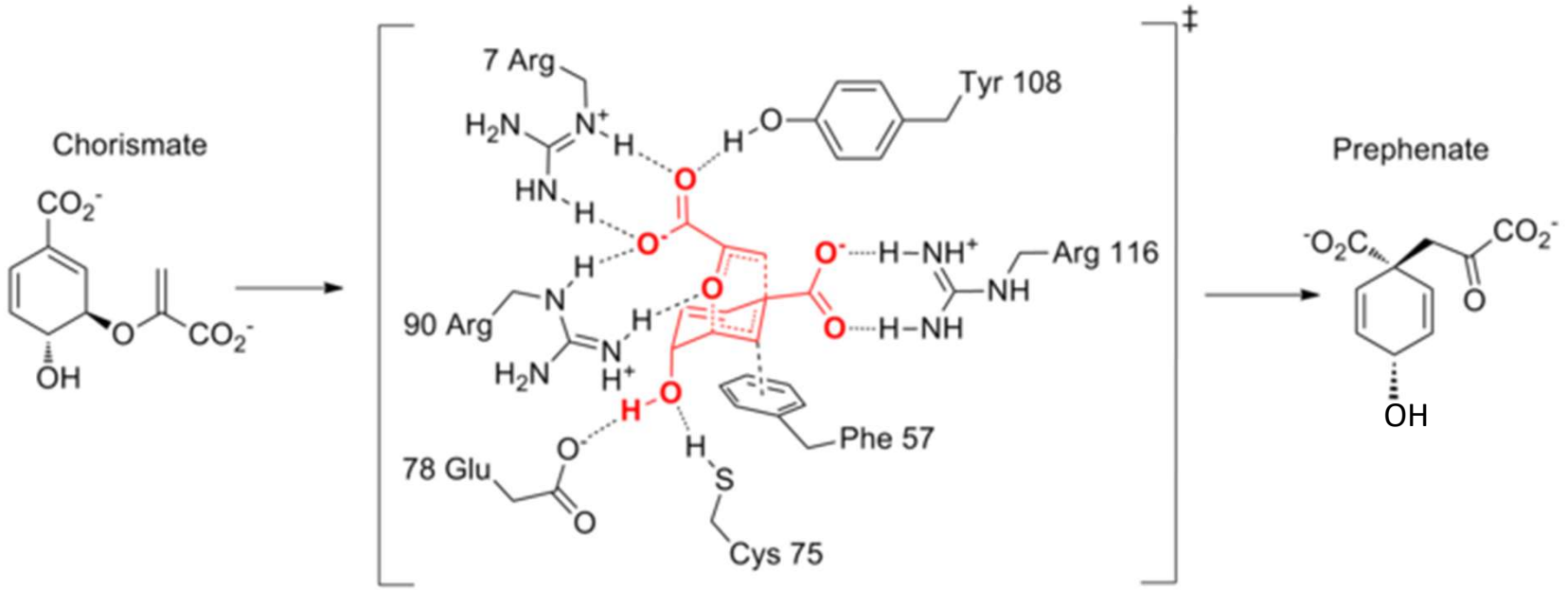
- Symmetrical homolytic bond breaking or homogenic bond making
- Bonds are broken and formed by reaction of species that have an odd number of electrons (radical intermediates)

**(3) Pericyclic mechanism**

- Electrons move in a closed ring
- No intermediate is involved

# Pericyclic reactions (features)

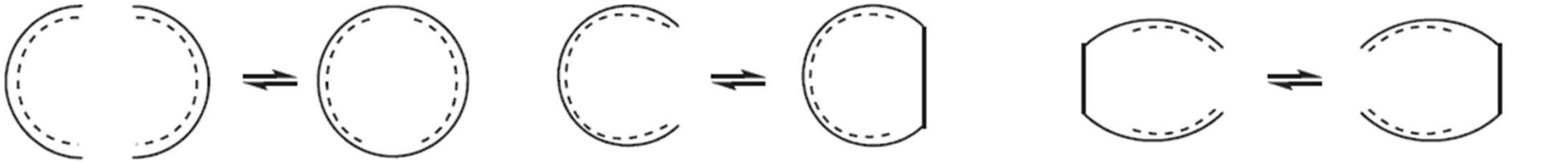
- Transition state possesses a cyclic geometry
- No new radical or ionic intermediate within the cyclic transition state
- Bond-making and bond-breaking progress in a concerted fashion
- Stereospecific and/or stereoselective




Chorismate mutase catalyzes the Claisen rearrangement of chorismate

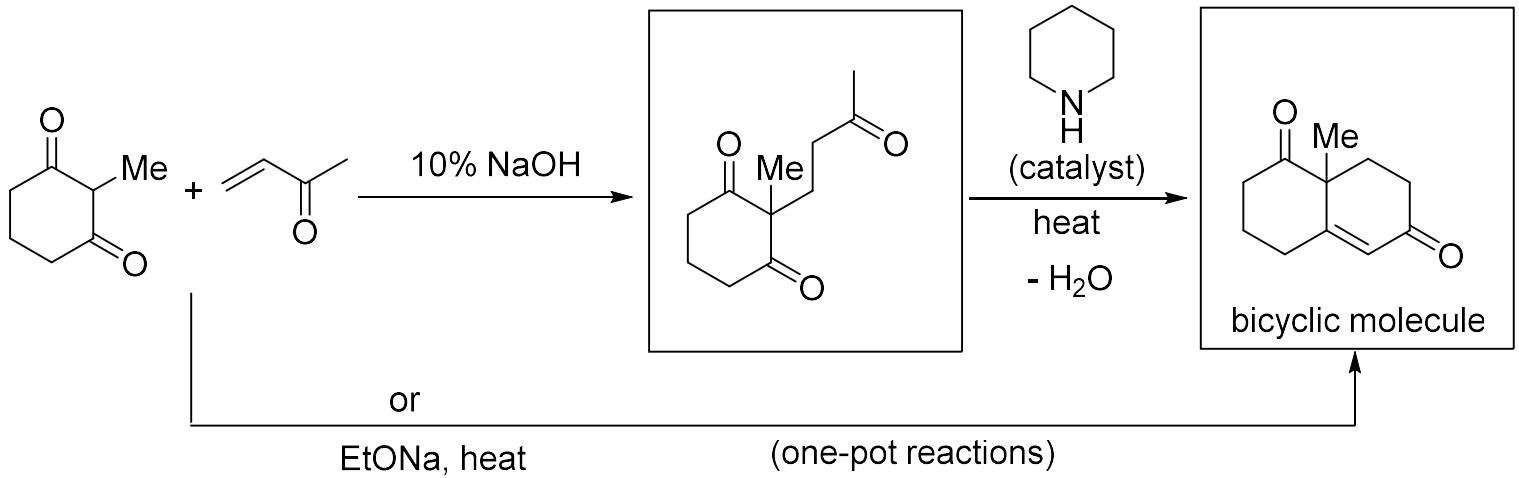
# Pericyclic reactions (classifications)

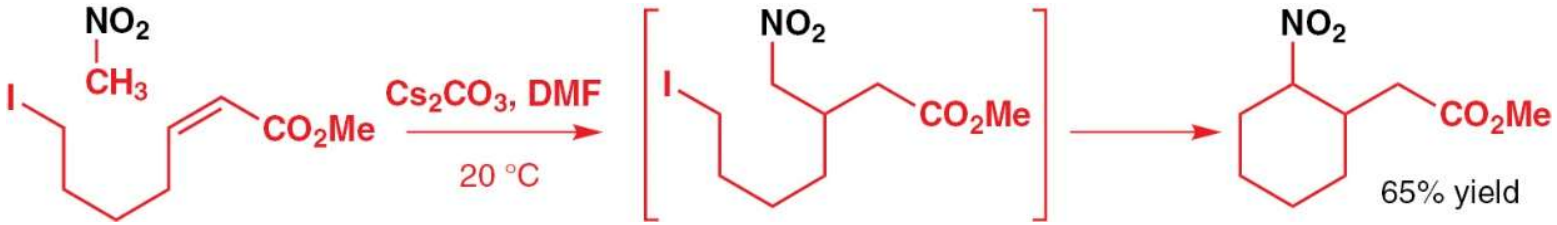
- Cycloadditions
  - Diels-Alder reaction
  - 1,3-dipolar cycloadditions
  - Other types of cycloadditions
- Electrocyclic reactions
- Sigmatropic rearrangements



(a) Cycloadditions (b) Electrocyclic reactions (c) Sigmatropic rearrangements

# Examples of formation of six-membered rings (polar mechanisms)

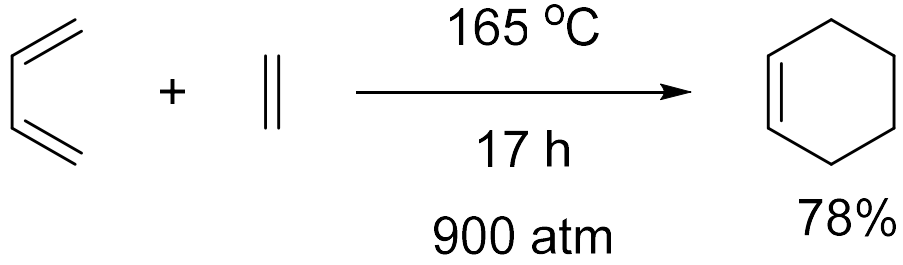


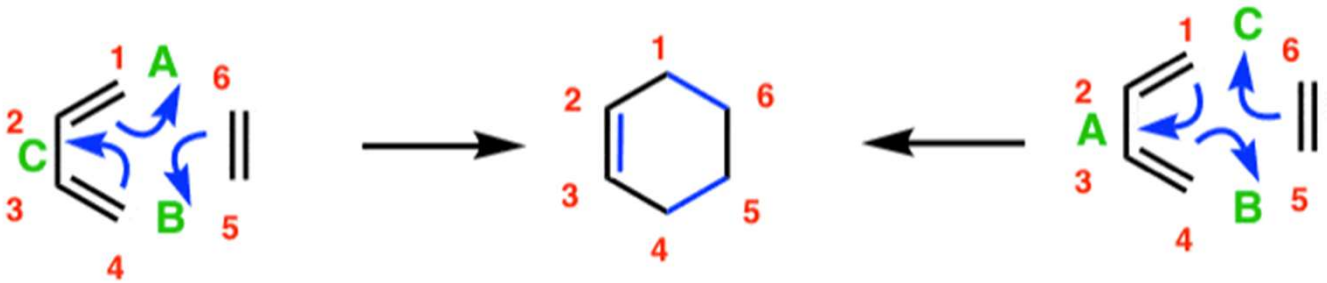



# 1. Describing Diels–Alder reaction

- Diels-Alder reaction is a reaction between a conjugated diene and an alkene to give cyclohexene.
- The most simple DA reaction is between butadiene and ethylene.
- First reported in 1928, Nobel prize in 1950

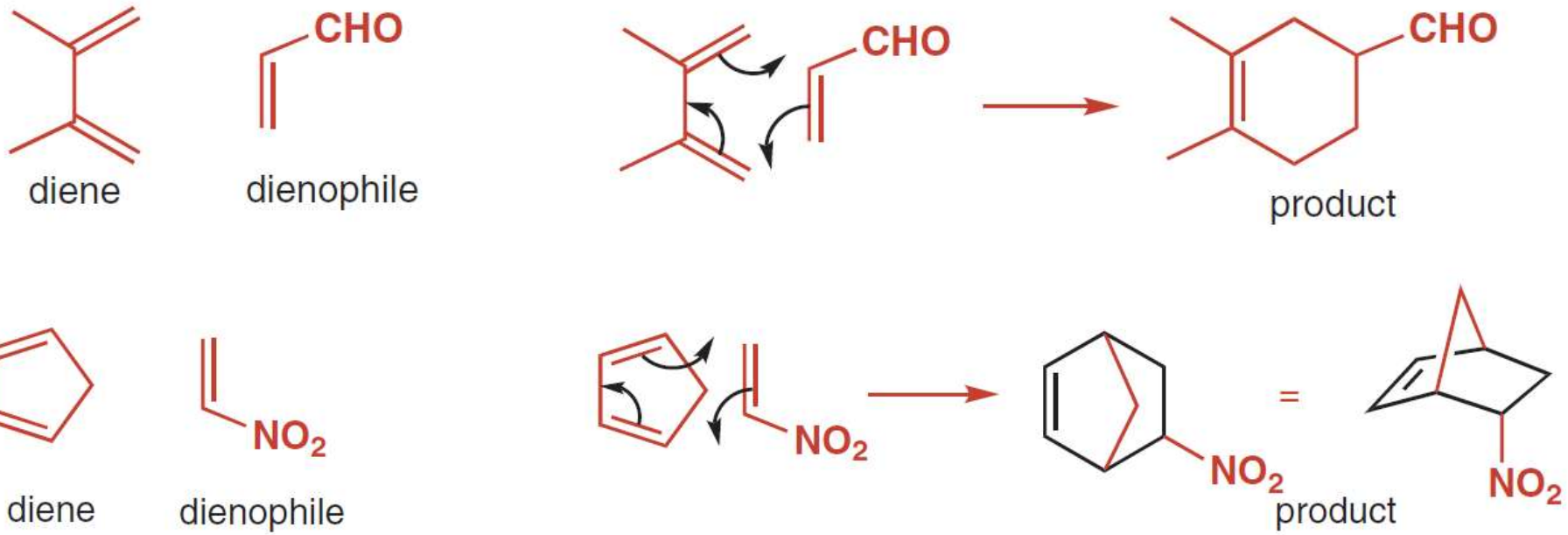




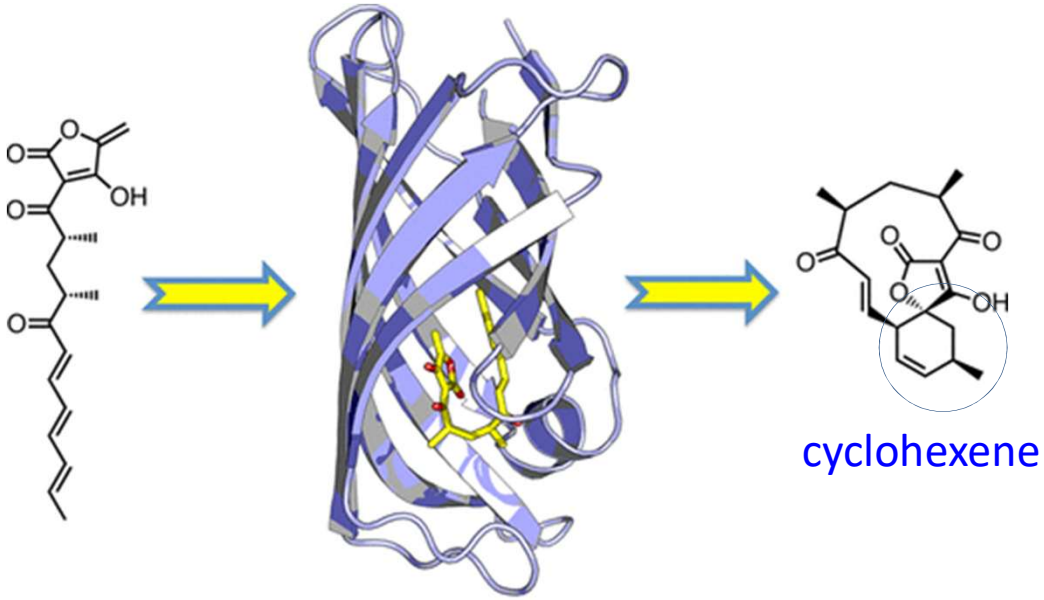



# Reactants and products

- Conjugated diene: 2 C=C double bonds connected by 1 single bond.
- Alkene (sometimes alkyne) is typically refer to as dienophile.
- Product: cyclohexene



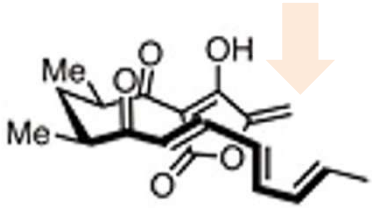
# Diels–Alder reactions in nature




cyclohexene

Diels-Alderase catalyzed [4+2] cycloaddition

dienophile

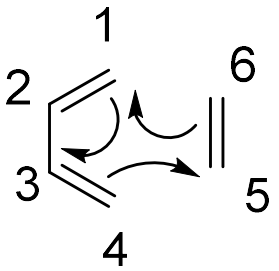



diene

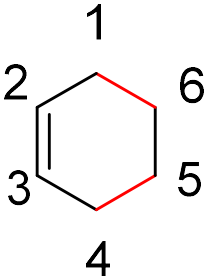
*J. Am. Chem. Soc.* 2016, 138, 19, 6095–6098

# 2. Mechanism

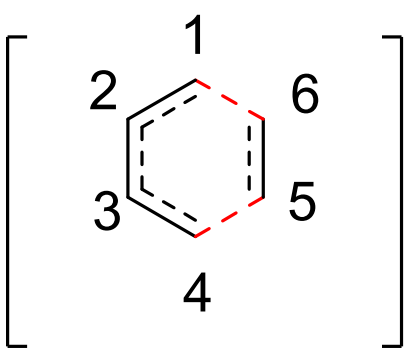
- A type of pericyclic reaction:
  - Single elementary step,
  - Electrons move round a circle,
  - No charge building up and no intermediates,
  - All the bond forming and breaking occur at the same time
- Thermodynamics: two C-C sigma bonds are formed, C1-C6 and C4-C5, at the expense of two C-C $\pi$ bond.
- Kinetics: six delocalized $\pi$ electrons and thus is aromatic (4n+2) in character (stabilization of transition state)




Arrows can be draw clockwise or anti-clockwise

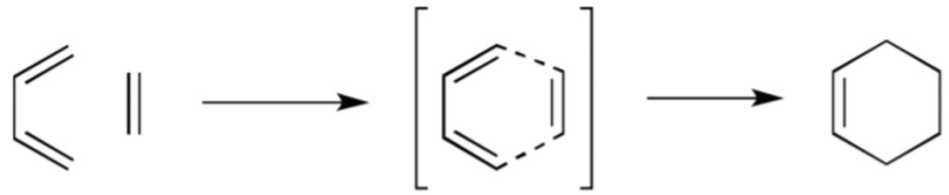


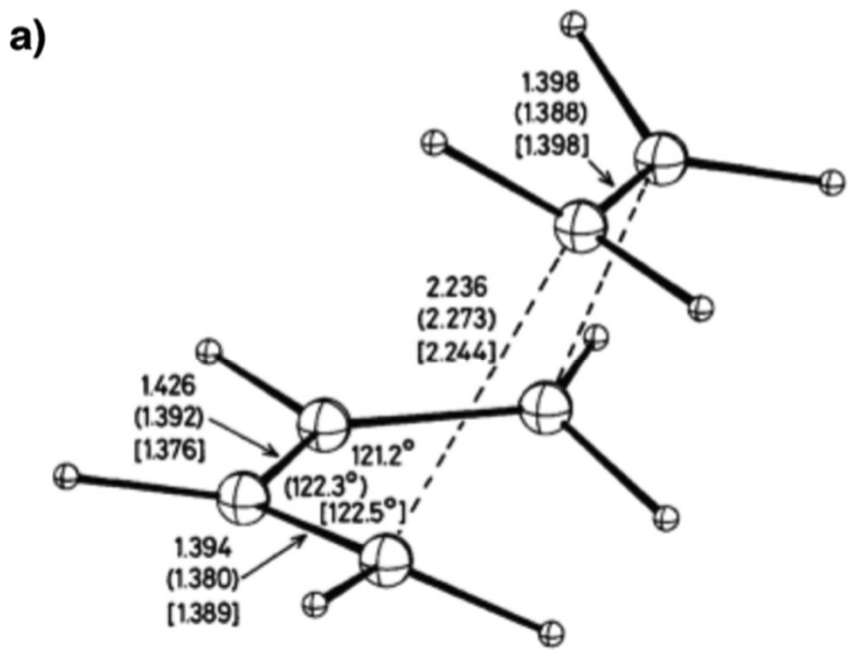

2 new C-C sigma bonds C1-C6 and C4-C5

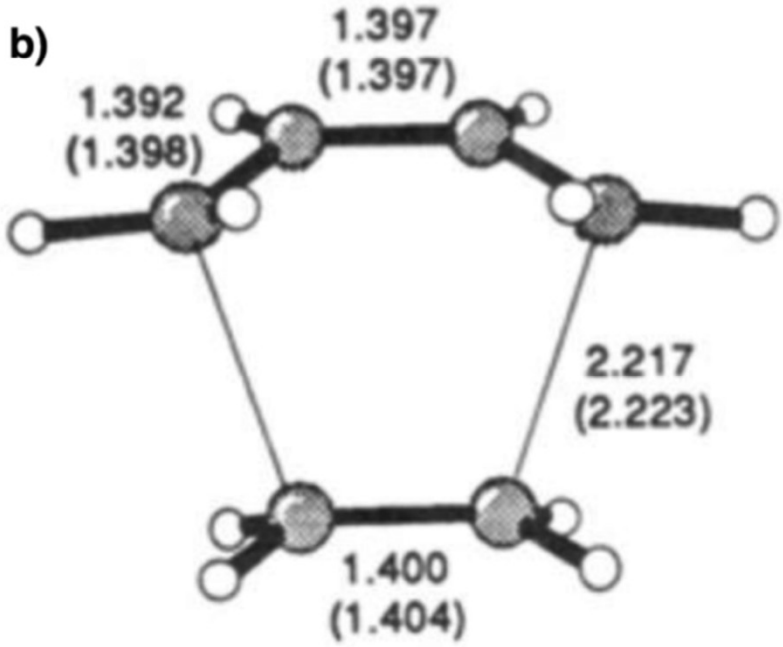



transition state

# Transition state (computational studies)

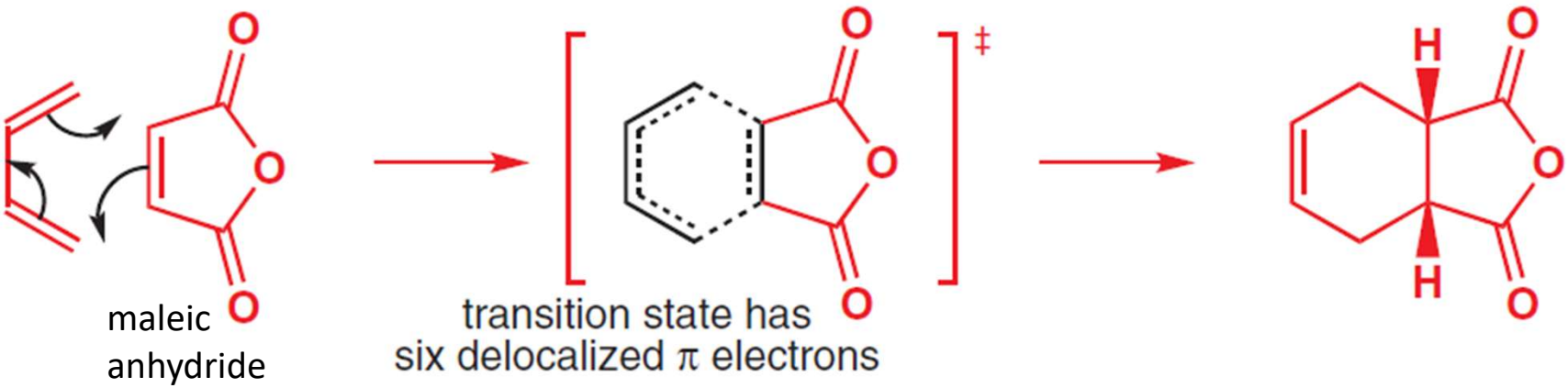




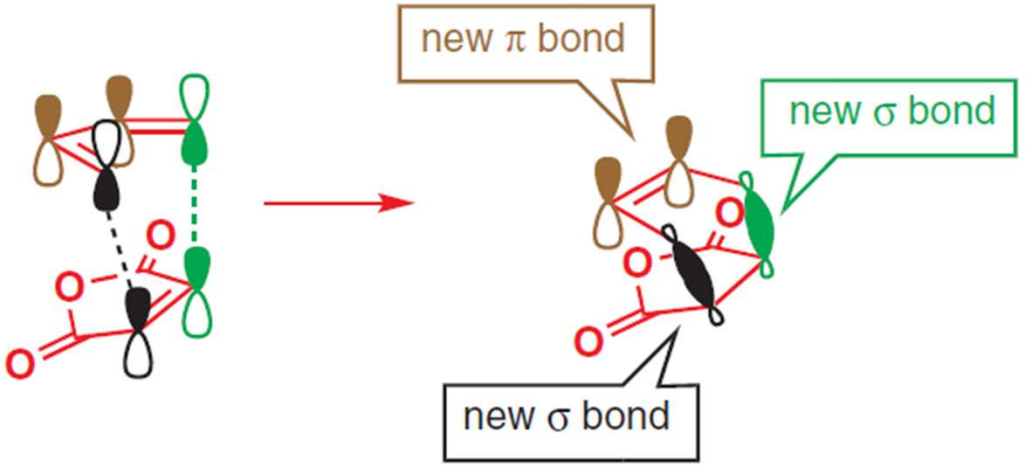



# Transition state

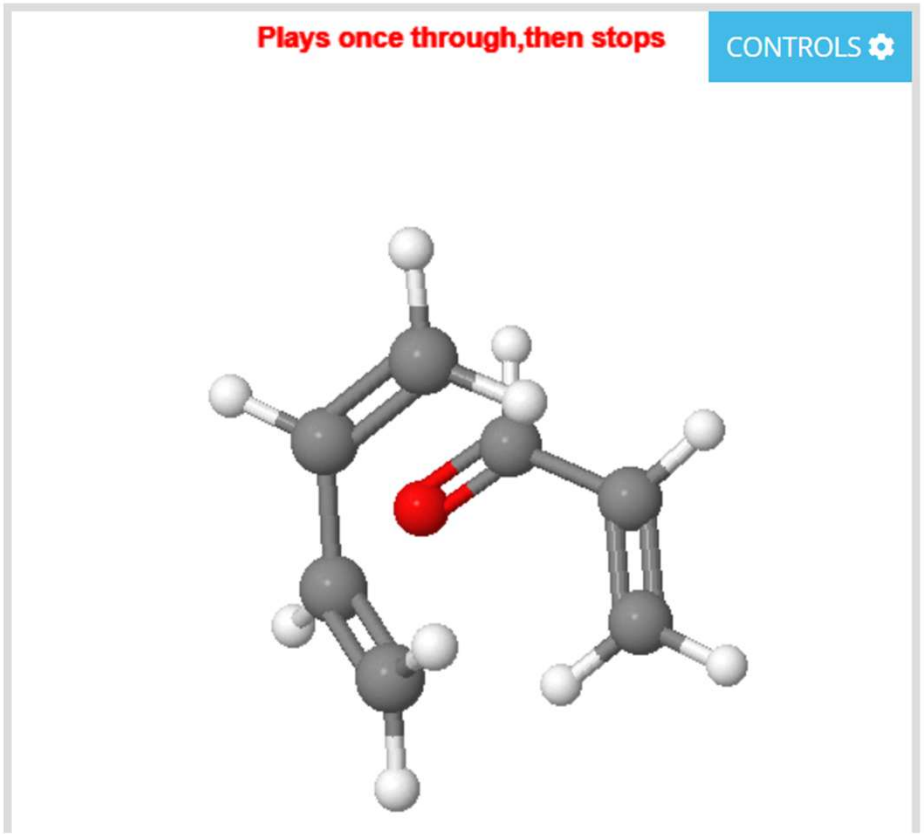
**6 delocalized π–electrons**




**suprafacial-suprafacial approach**

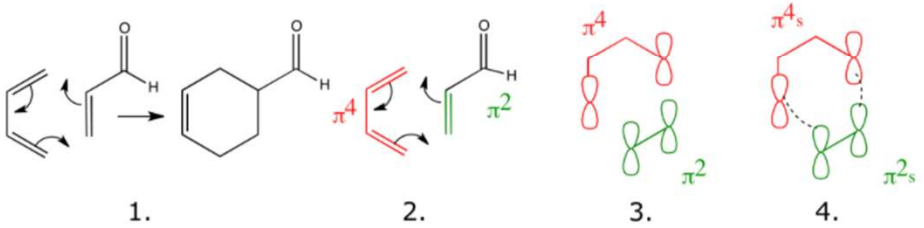


## Thermal [4 + 2]




The best way to make sense of the Woodward-Hoffmann rules is to try using them.

Click on the numbers and structures below to see how this works for the Diels-Alder reaction.




'In allowed thermal pericyclic reactions the number of $(4q+2)_s$ and $(4r)_a$ components must be odd.'

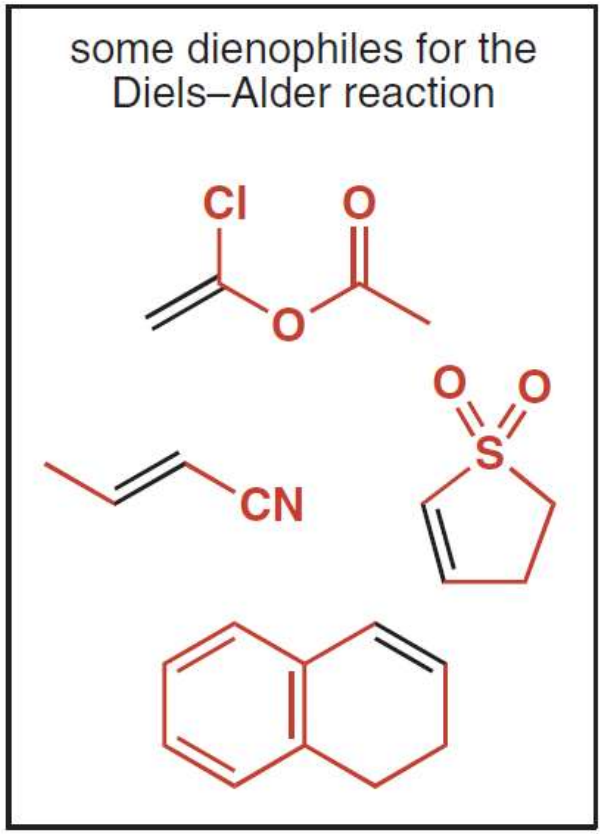
Let's break it down:

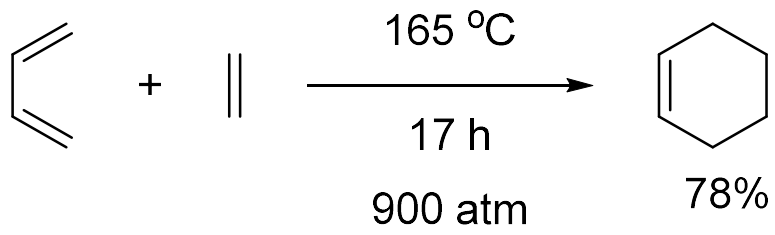
A **component** is a **bond or orbital** that takes part in the reaction as **one unit**. Both the **diene** and the **dienophile** are **each one single component**. We label the components by the **type** and **number of electrons** there are

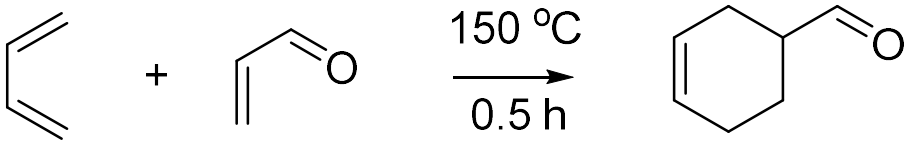
https://www.chemtube3d.com/dawoodward-hoffman/

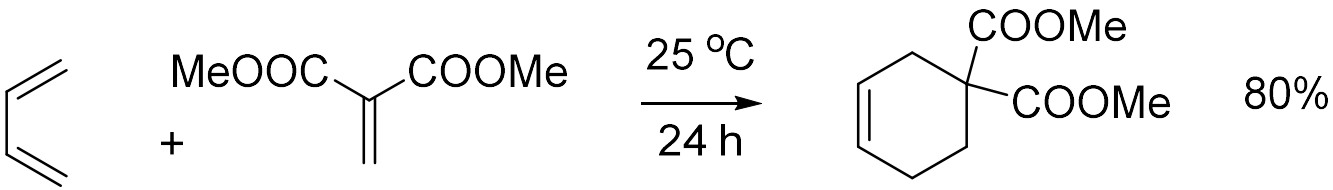
# 3. Reaction rates: dienophiles

- It is common that the dienophile contains EWG (electron withdrawing groups, $NO_2$, CN, $SO_2R$, COOR, COR, Cl, Ph)
- Typically, the more electron deficient the dienophile is, the faster the reaction.



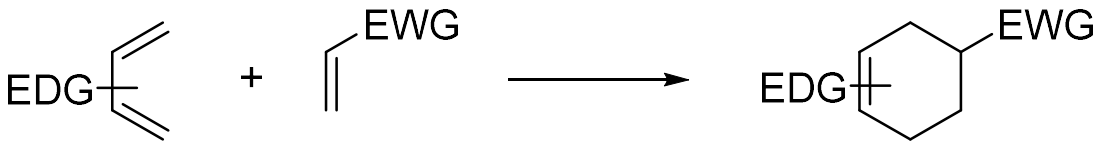



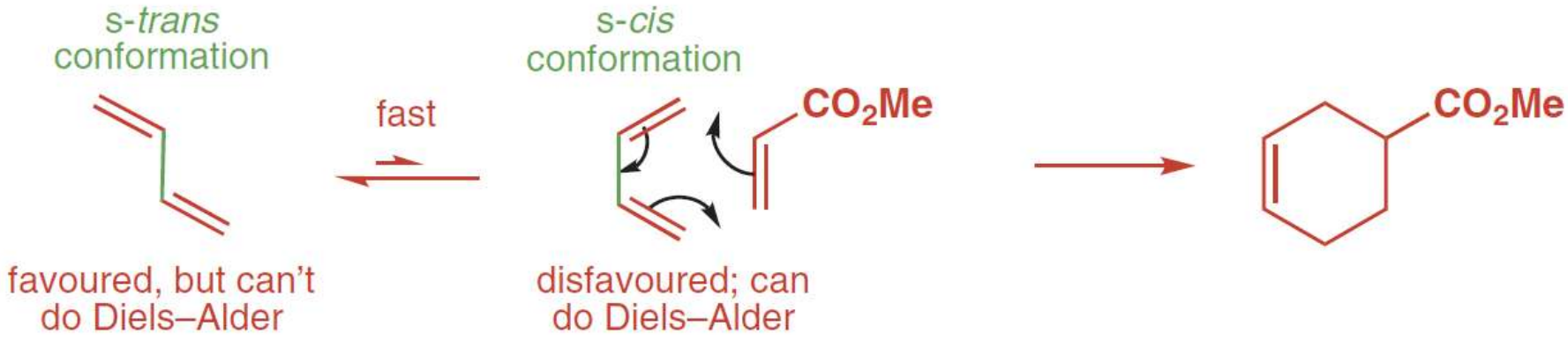



# Dienes

- More electron rich diene make DA reaction faster than electron-neutral or electron-deficient dienes.



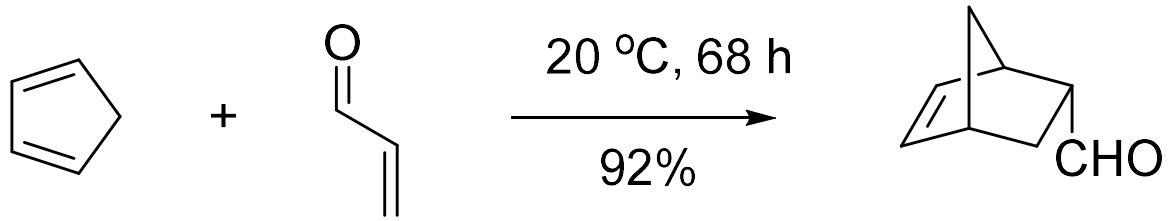

- For the diene to react, it must take up a conformation where C1 and C4 are close together (*s*-cis conformation)



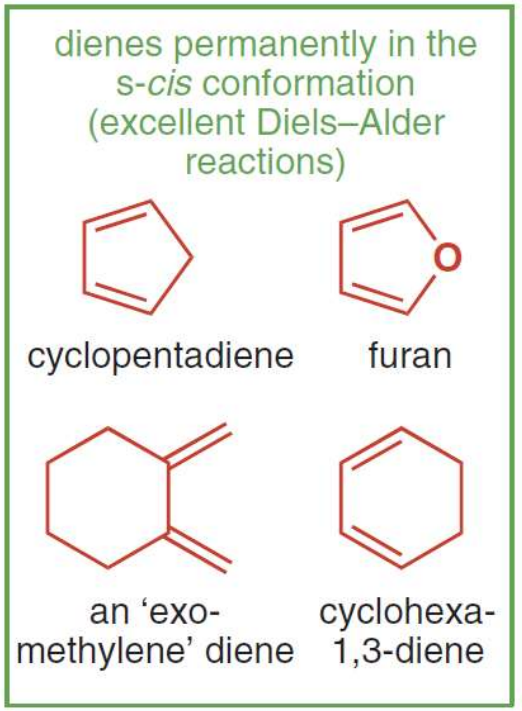

For butadiene: only 1% of its molecules at rt in the s-cis conformation

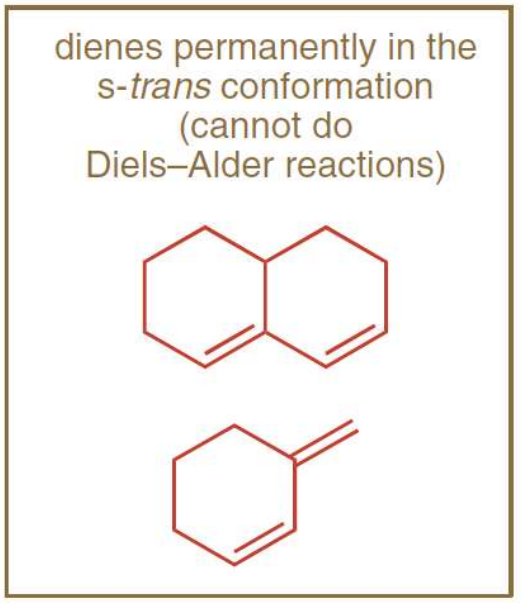
# Cyclic dienes

- Cyclic dienes (cyclopentadiene, furan etc) are locked at s-cis conformation. They are significantly more reactive than open chain dienes and are good substrates for DA

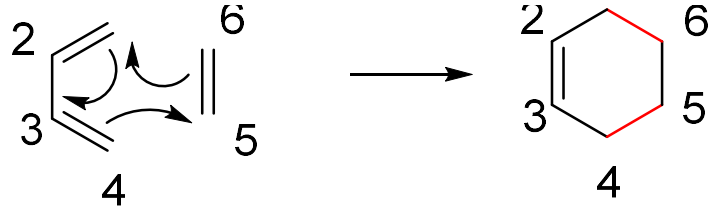


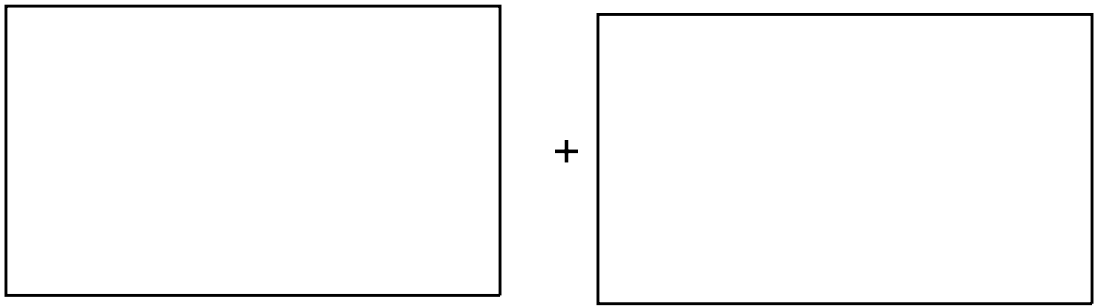

- Dienes that cannot adopt *s*-cis conformation cannot participate in DA reaction.

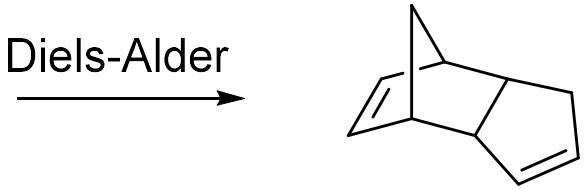


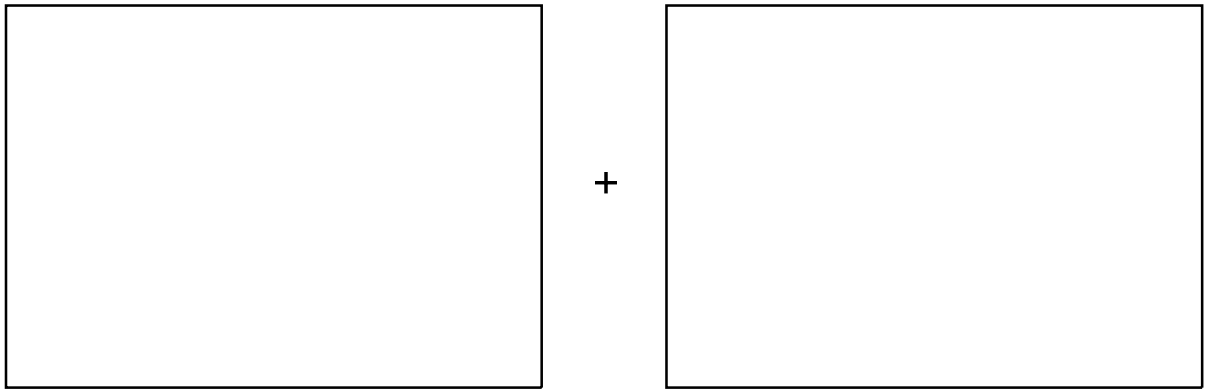



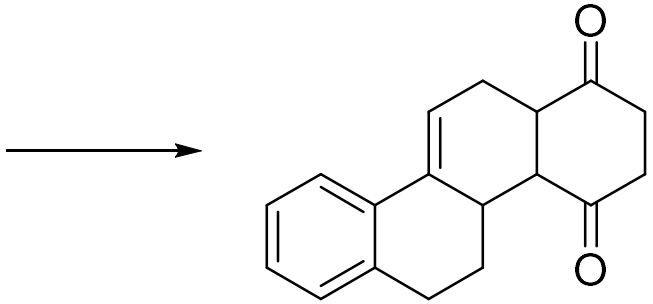
# Practice



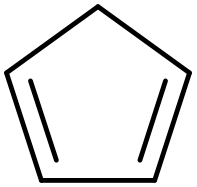





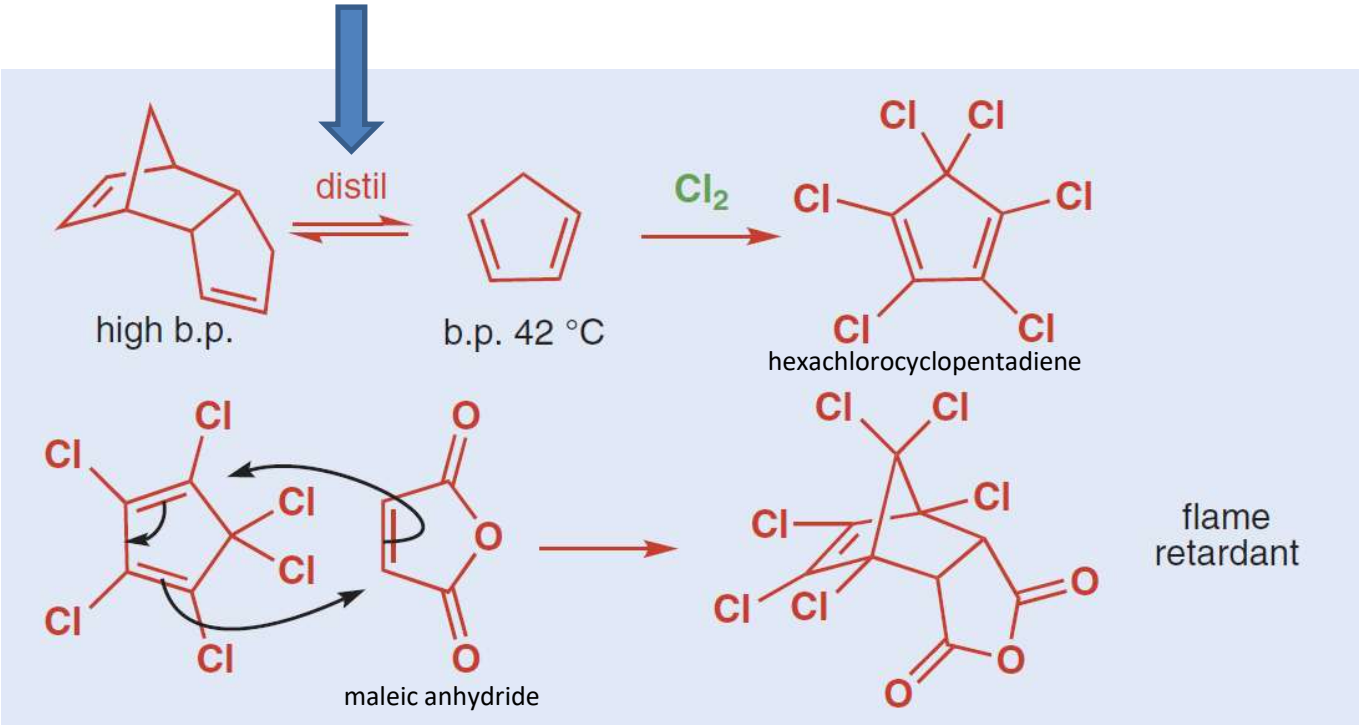





# Cyclopentadiene



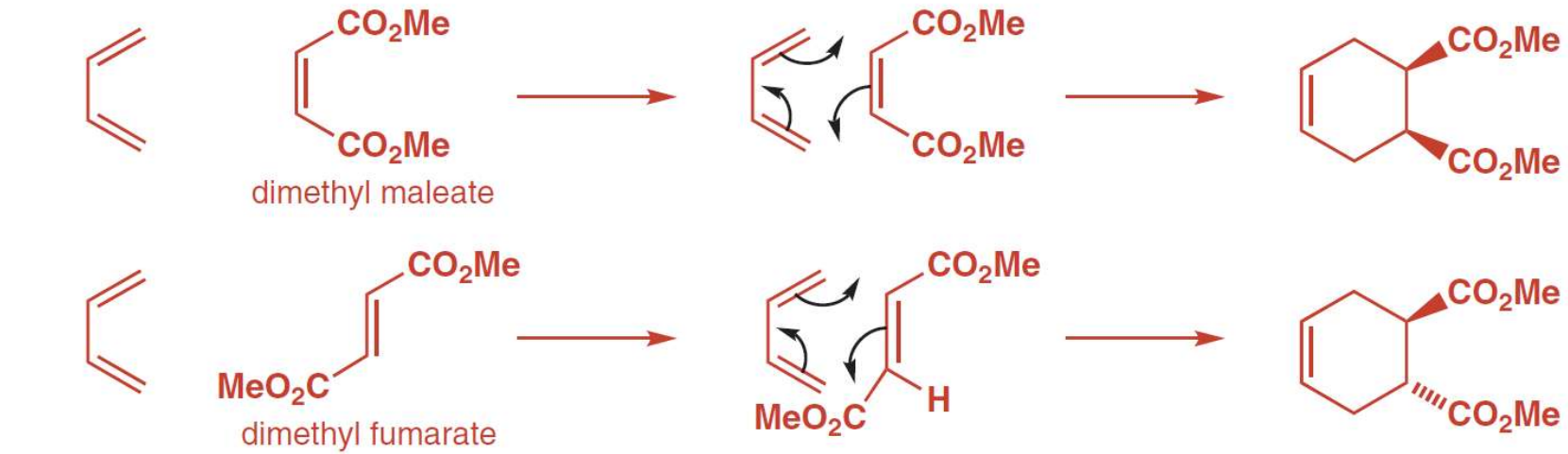
- At room temperature, it exists as dimer form. To use cyclopentadiene, one must heat the dimer to crack it.
- The monomer can be stored for days at −20 °C, but will dimerize again at room temp within a few hours.
- Diels-Alder reaction could be reversible.



# 4. Stereospecificity: bis-substituted dienophile

- Dienophile:

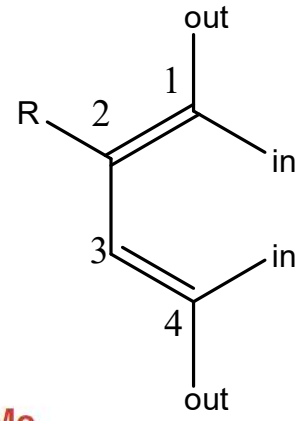
cis dienophile → cis product, trans dienophile → trans product

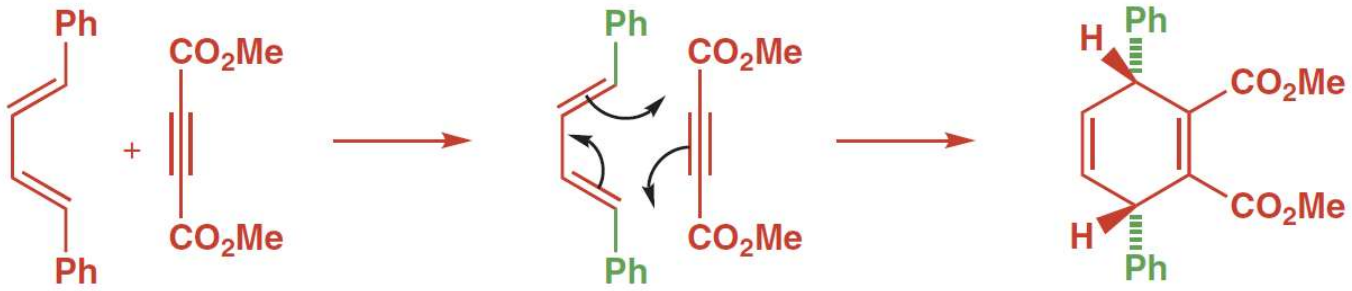


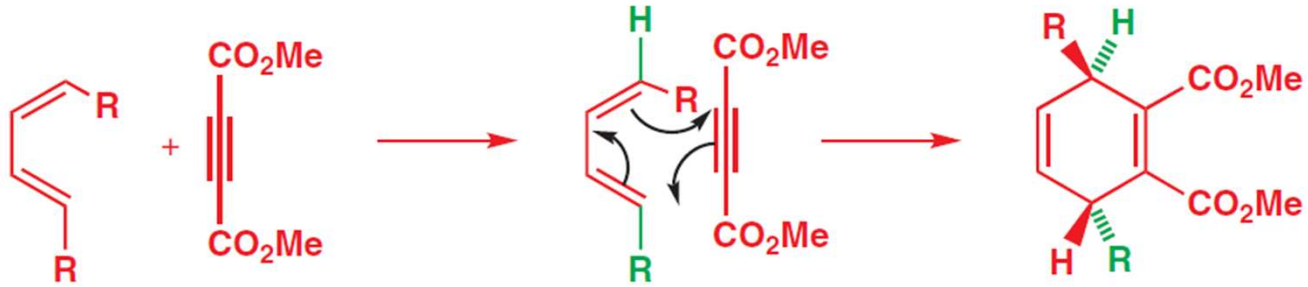
# Stereospecificity: bis-substituted diene

Diene: “in” and “out” substituents in the s-cis conformation

- Two “in” substituents are 1,4-cis in the product
- Two “out” substituents are 1,4-cis to each other

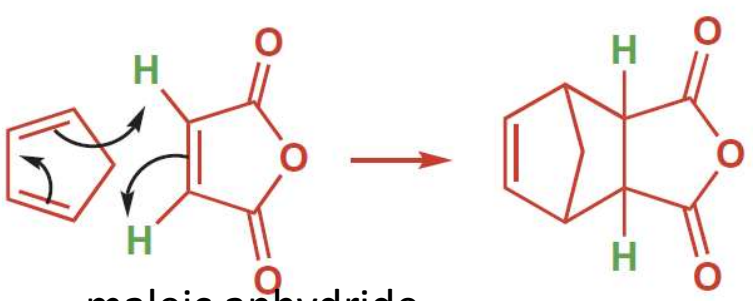


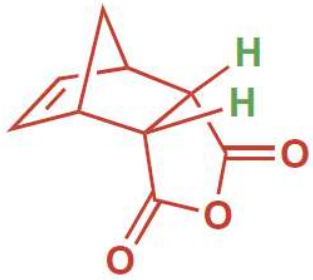



# Stereoselectivity and the endo rule: substituted dienophile and substituted diene

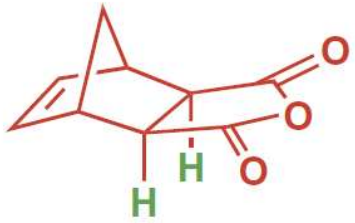
Diels Alder reaction typically favors endo-transition state




maleic anhydride

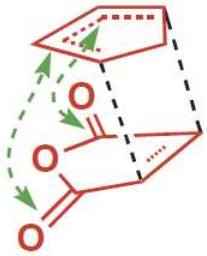



the *endo* adduct (formed)

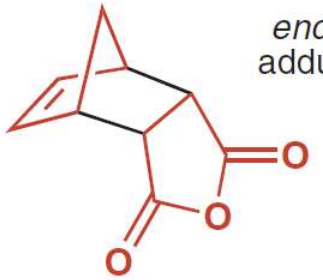



the *exo* adduct (not formed)

bonding interaction in transition state between C=O groups and back of diene




new double bond and C=O groups end up on same side of molecule: *endo*



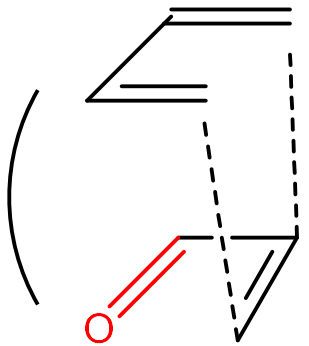

*endo* adduct

secondary orbital overlap in endo-transition state

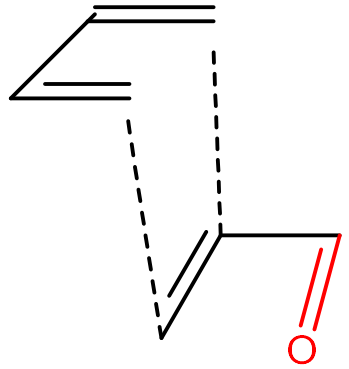
# Endo and exo transition states

The diene and the dienophile are both planar and arranged on parallel planes in the transition state. The EWG is placed right below the diene (endo transition state).

EWG below the diene




endo transition state

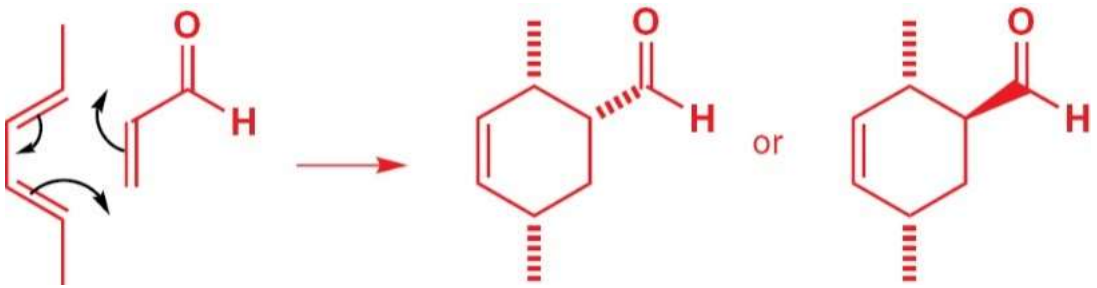



exo transition state

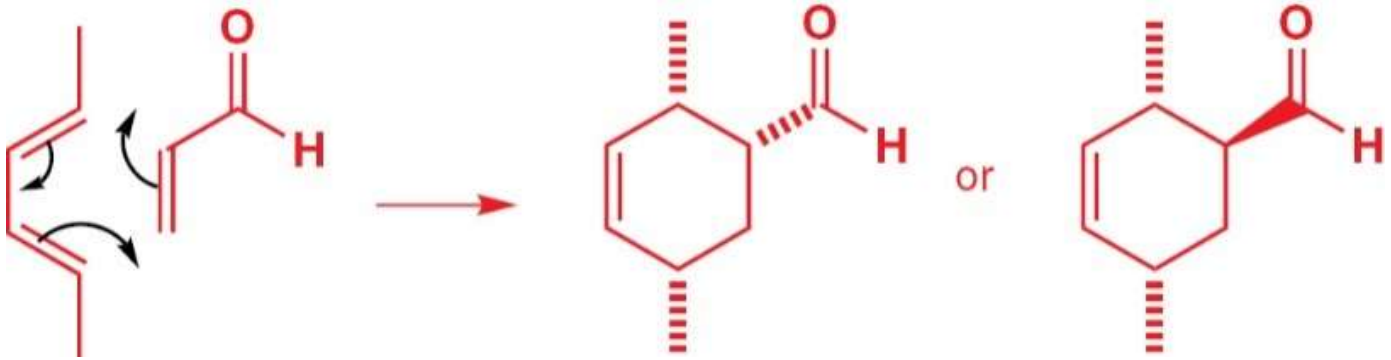
- The endo mode of addition is usually preferred when an substituent such as a carbonyl group is present on electron-withdrawing the dienophile.
- Frequently a mixture of both stereoisomers is formed.

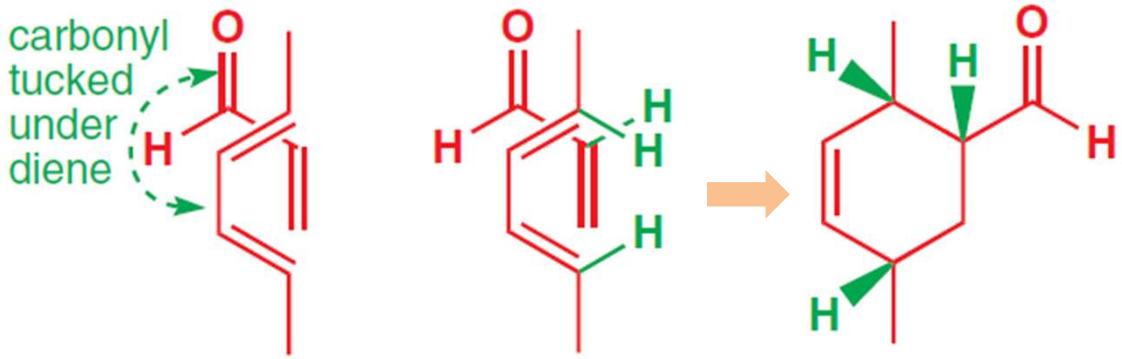
# Practice

**Which product is favored according to the endo rule?**



**Which product is favored according to the endo rule?**

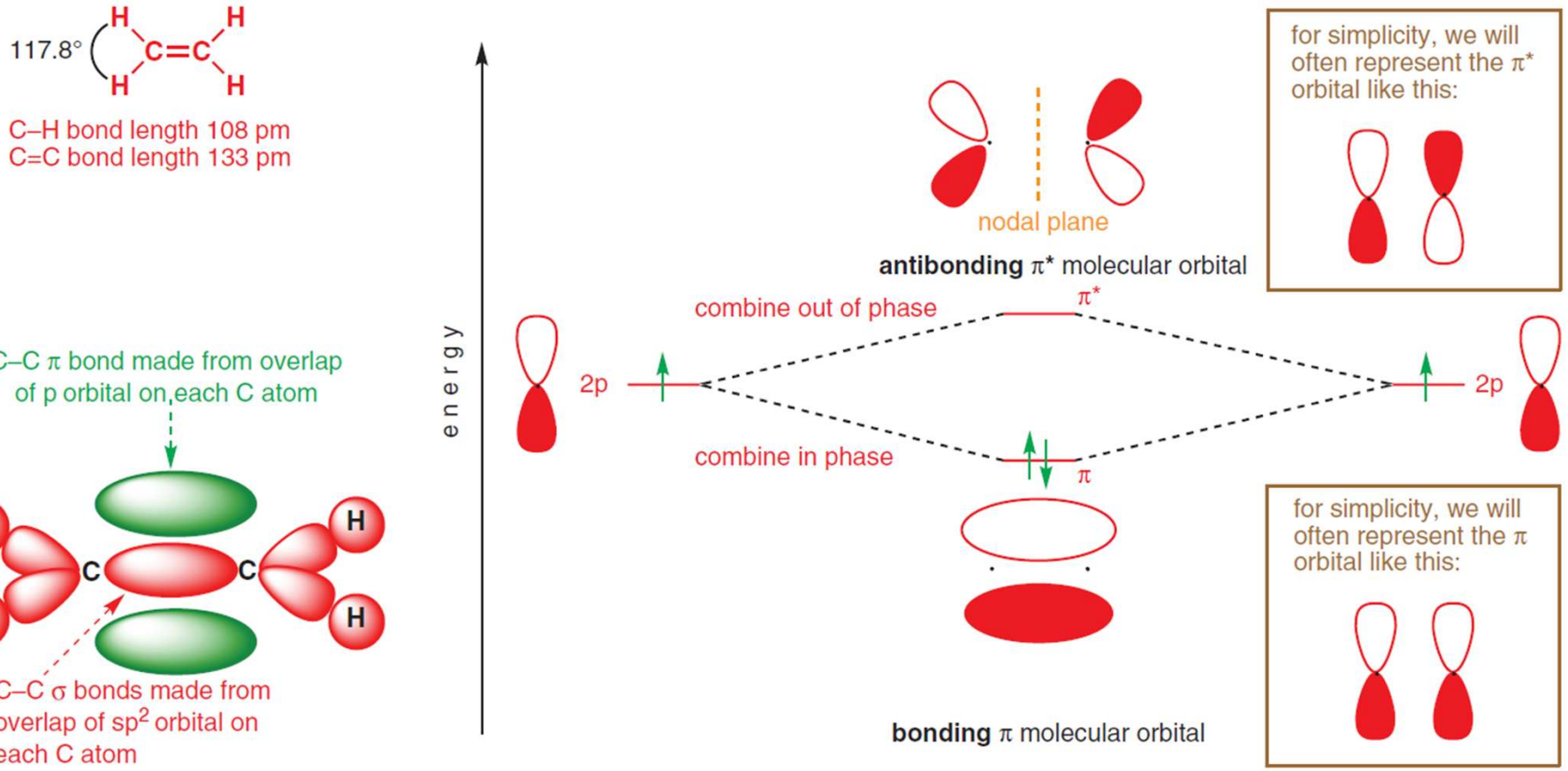


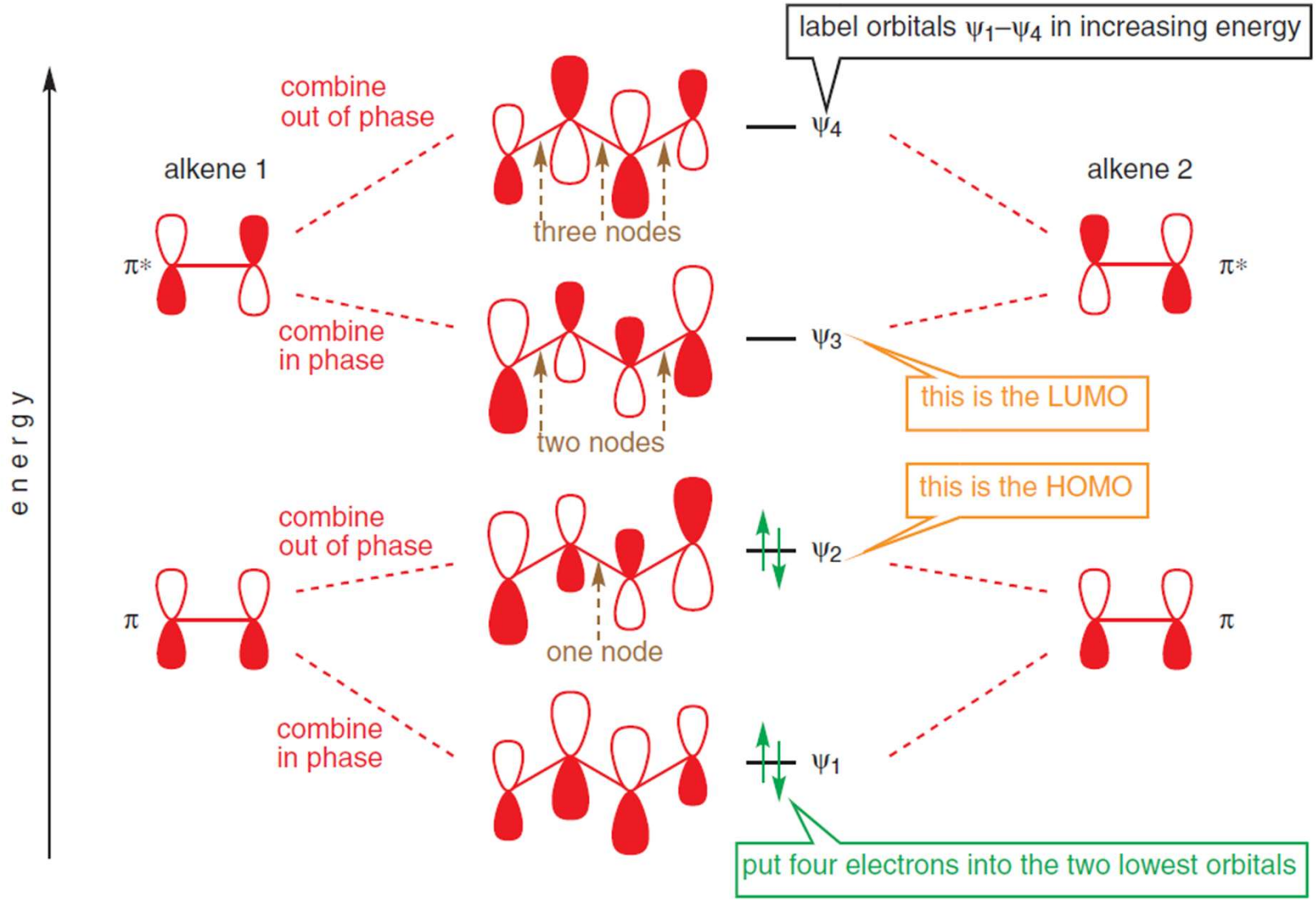

endo transition state

endo product

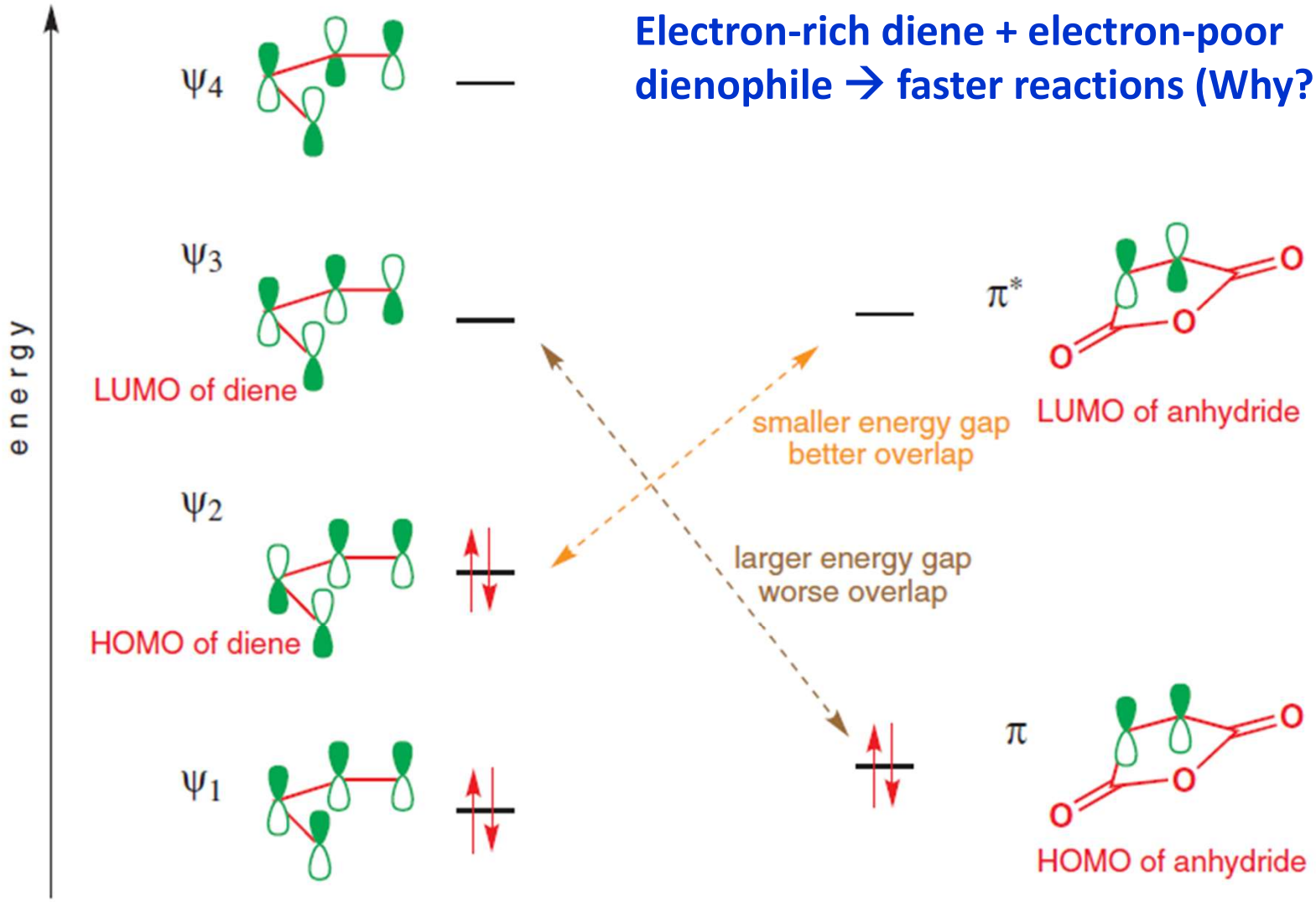
# 5. Molecular orbitals of alkene



# Molecular orbitals of diene

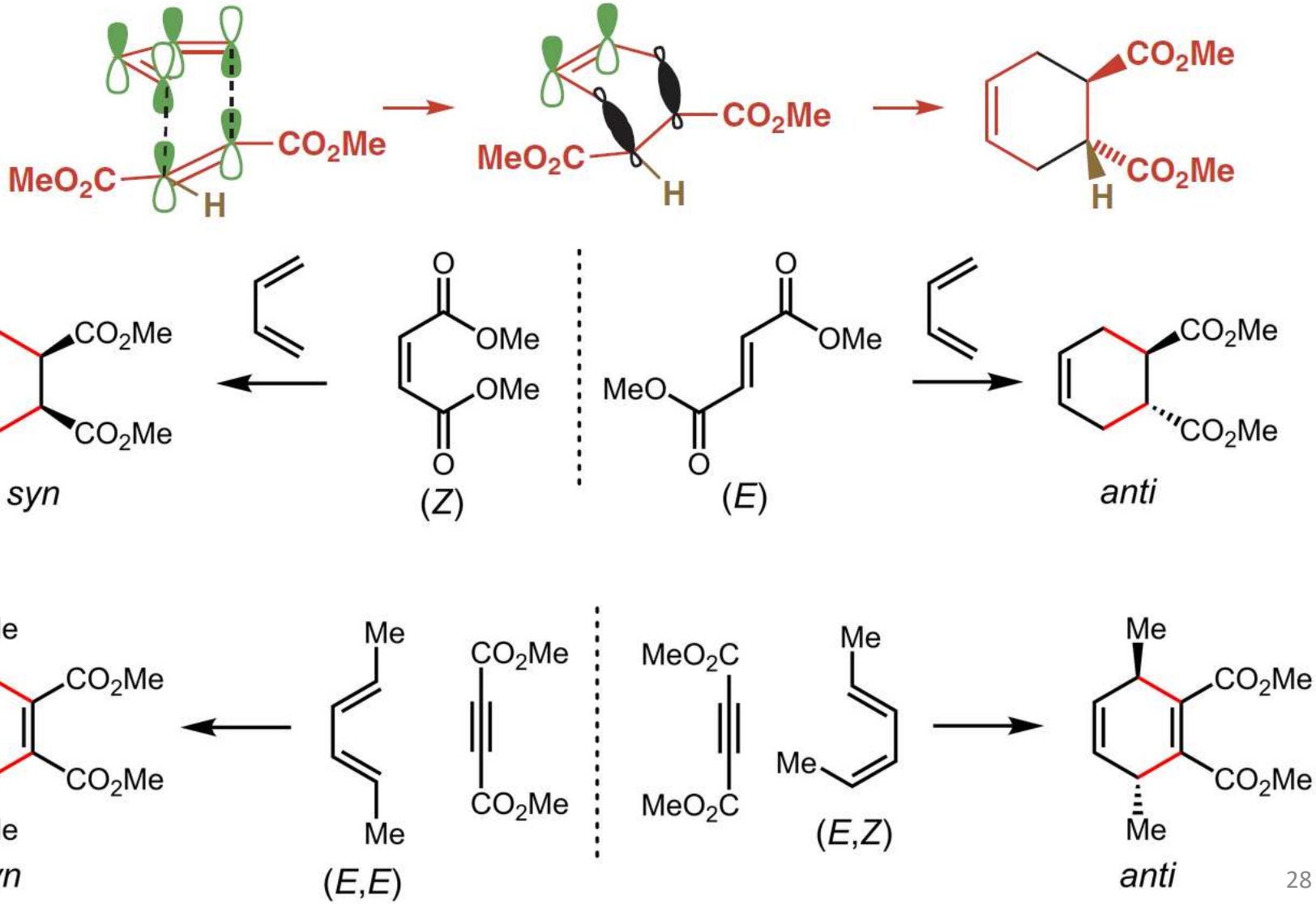


# The frontier orbitals of DA reaction

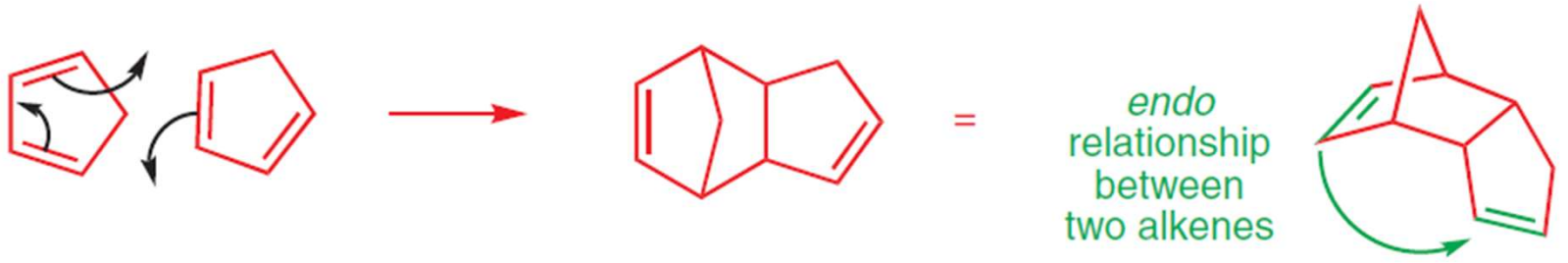



**Electron-rich diene + electron-poor dienophile → faster reactions (Why?)**

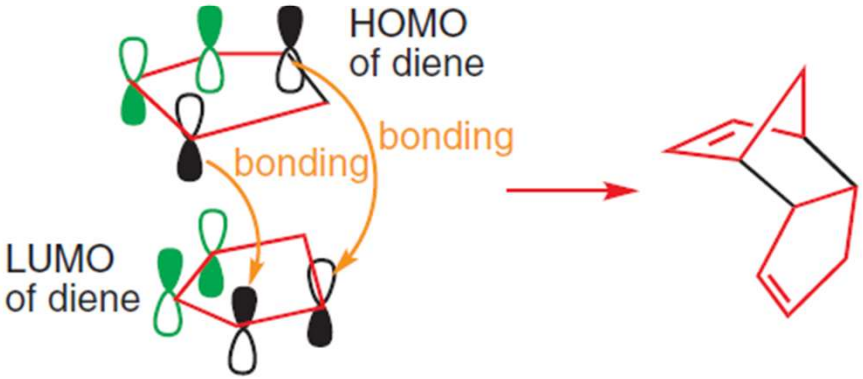
# Frontier orbitals and Stereospecificity



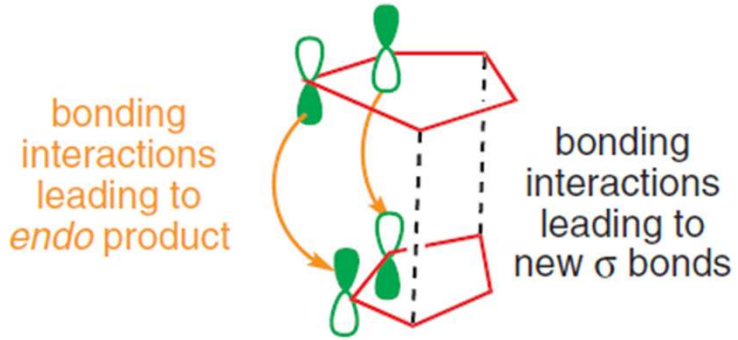
# Frontier orbitals and preference for endo TS in Diels–Alder reactions




orbital interactions leading to new bonds



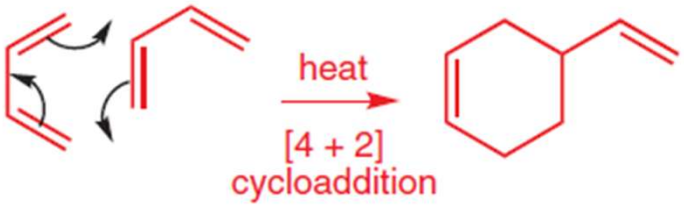

orbital interactions favouring endo product

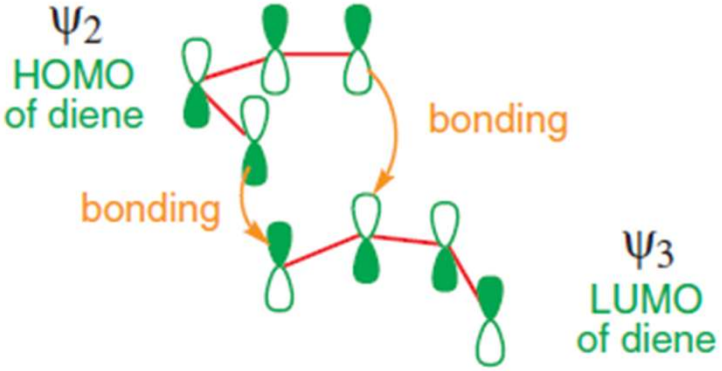


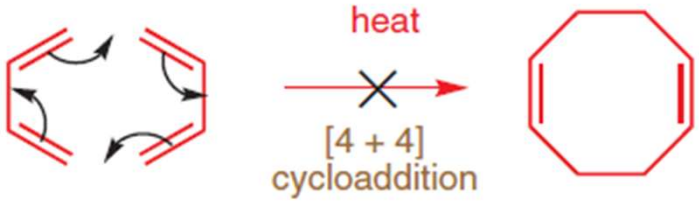

secondary orbital overlap
in endo-transition state

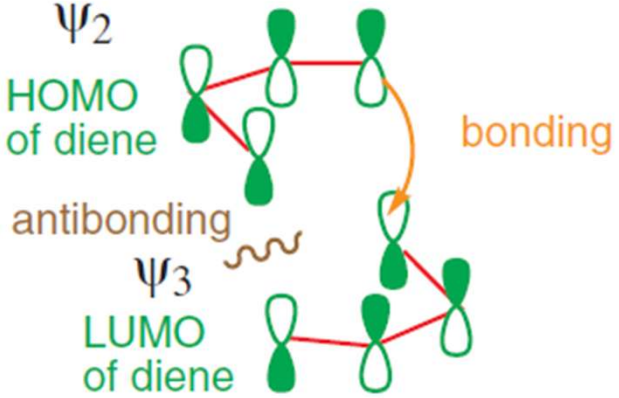
# The frontier orbitals of DA reaction

**1,3-butadiene undergoes thermal [4+2] reaction, but not thermal [4+4] reaction, Why?**



# Summary

- Diels-Alder reaction
  - A reaction between a conjugated diene and an alkene to give cyclohexene
- Mechanism of Diels–Alder reaction
  - Concerted mechanism; cyclic transition state
- Kinetics of Diels–Alder reaction
  - Electron rich diene + electron deficient dienophile → faster rate
- Stereochemistry of Diels–Alder reaction
  - Stereospecificity for substituents within each reactant
  - Stereoselectivity between substituents of the two reactants (endo TS)
- Frontier molecular orbitals and Woodward–Hoffmann rules
  - Explanation of stereochemistry using FMOs
  - Prediction reactivity using FMOs and Woodward–Hoffmann rules

# 6. Woodward–Hoffmann rules

Harvard University DEPARTMENT OF CHEMISTRY

R. B. WOODWARD • 12 OXFORD STREET • CAMBRIDGE • MASSACHUSETTS 02138

November 25, 1964

Dr. Marshall Gates
Department of Chemistry
University of Rochester
Rochester 20, New York

Dear Marshall:

Dr. Hoffmann and I feel that the Journal is unquestionably the place for the accompanying communication, which puts forward a simple and important general principle, applicable to a very wide range of phenomena of great current interest. We are concerned about its length--more properly, we feel that you will be concerned about its length--but we hope you will see your way to including it. If you cannot, just send it back, and we will proceed to present it for publication in a less satisfactory vehicle.

I'll ring you up Friday or Monday to discuss it further with you.

Best personal regards and all good wishes, as ever.

Yours,

R. B. Woodward

RBW:DD
Enclosures

JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
BOARD OF EDITORS

Department of Chemistry
University of Rochester
Rochester, New York 14627

Dec 1 1964

Dear Sir:

Your paper entitled 4073 COMM ED: Stereochemistry of Electrocyclic Reactions has been accepted for publication and will probably appear in Jan1965. You should receive galley proof at least two weeks prior to the date of publication. Revised galley or page proof is not sent to authors unless requested. For the convenience of the Editor in correcting errors, please return the manuscript when you return the proof. Reprint order form is sent with proof.

MARSHALL GATES, *Editor*

## The Nobel Prize in Chemistry 1981



Photo from the Nobel Foundation archive.
Kenichi Fukui
Prize share: 1/2



Photo from the Nobel Foundation archive.
Roald Hoffmann
Prize share: 1/2

The Nobel Prize in Chemistry 1981 was awarded jointly to Kenichi Fukui and Roald Hoffmann "for their theories, developed independently, concerning the course of chemical reactions."

J. Org. Chem. 2015, 80, 11632−11671

# Woodward–Hoffmann rules

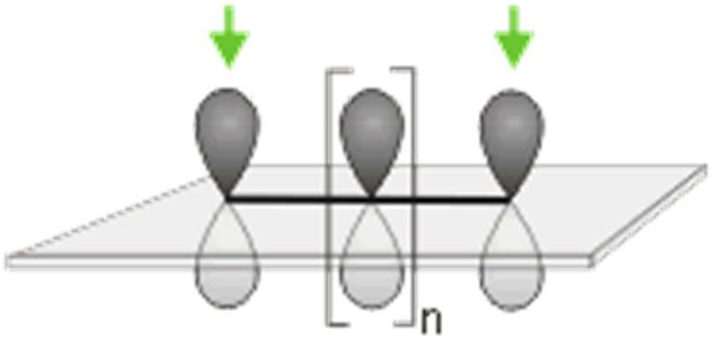
- **In a thermal pericyclic reaction the total number of (4q + 2)s and (4r)a components must be odd.**

In an allowed thermal pericyclic reaction this sum:

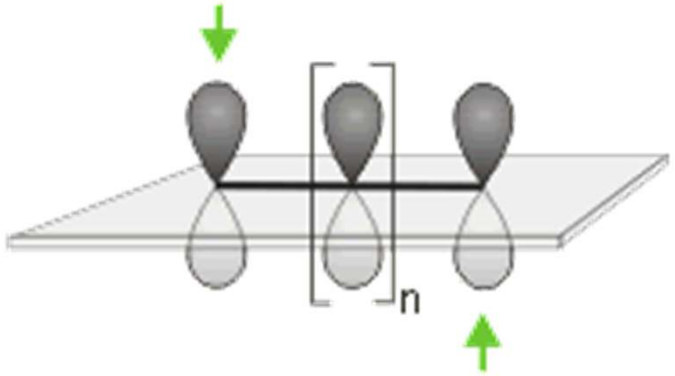
number of suprafacial components with 2, 6, or 10 electrons
\+ number of antarafacial components with 0, 4, or 8 electrons
= an odd number

'Allowed' doesn't mean that it will happen.
It just means it is theoretically possible.

# Woodward–Hoffmann rules




Suprafacial

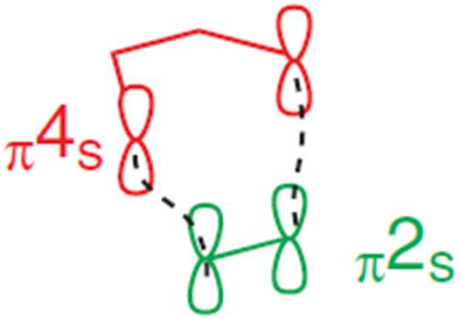



Antarafacial

A **suprafacial** component forms new bonds on the same face at both ends.

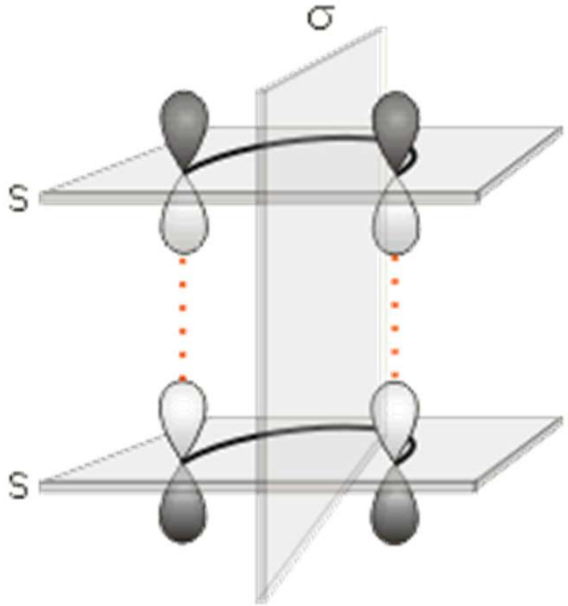
An **antarafacial** component forms new bonds on opposite faces at both ends.

# Woodward–Hoffmann rules




**DA reactions:**

- **1 * (4q + 2)s component (the alkene)**
- **0 * (4r)a component.**
- **Total = 1 so it is an allowed reaction.**




suprafacial-suprafacial approach

# Practice

**Use Woodward–Hoffmann rules to explain: 1,3-butadiene undergoes thermal [4+2] reaction, but not thermal [4+4] reaction, assuming suprafacial-suprafacial approach**

